BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

Course Code: CSE 219 Course Title: Signals and Linear Systems

Time: 25 minutes **Marks: 20**

Student Name: ________________________________ Student ID: ____________

Answer **all of Questions 1–3**. Answer concisely.

1. Consider the sequence
$$x[n] = \{2, 0, 1, -1\}, \qquad 0 \leq n < 4$$
You want to compute its 4-point DFT using the **radix-2 DIF FFT**. [8]
   (a) Perform the first butterfly stage and write the intermediate values.
   (b) Complete the second stage and write the outputs in the order produced directly by the DIF algorithm.
   (c) Rearrange the result from part (b) to obtain the DFT in natural order, i.e. $X[0], X[1], X[2], X[3]$.
   (d) Briefly state where bit-reversal appears in radix-2 DIT FFT and where it appears in radix-2 DIF FFT.

2. In an audio-processing application, a discrete-time signal $x[n]$ of length L is passed through a filter $h[n]$ of length M to produce
$$y[n] = (x * h)[n].$$
For large L and M, you want to compute this efficiently. [8]
   (a) What is the length of the output sequence $y[n]$? What is the time complexity of direct time-domain computation?
   (b) If you compute DFTs of insufficient length, multiply them, and then take an IDFT, what kind of convolution do you obtain instead of linear convolution?
   (c) Propose an FFT-based algorithm to compute the correct filtered output.
   (d) If the transform length is N, what condition must N satisfy? What is the resulting asymptotic complexity?

3. Thank you for attending the classes and participating in this CT. [4]